

Algebra 1
Unit 1
Lesson 10 Practice

Name **Answer Key** _____
Date _____
Period _____

1. To multiply two radicals with the same index, multiply the integers together and then multiply the radicands together. Then simplify the radical expression.

For questions 2-7, simplify each radical expression by writing an equivalent radical expression.

2. $\sqrt{11} \cdot \sqrt{11}$

$$\sqrt{121} = 11$$

3. $\frac{3\sqrt{3}}{-\sqrt{48}}$

$$\frac{3\sqrt{3}}{-\sqrt{16 \cdot 3}} \rightarrow \frac{3\sqrt{3}}{-4\sqrt{3}} = -\frac{3}{4}$$

4. $\sqrt[3]{8} \cdot \sqrt[3]{16}$

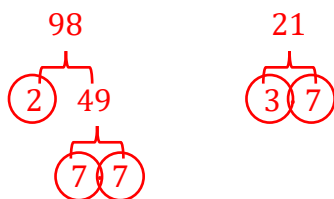
$$\frac{2 \cdot \sqrt[3]{8 \cdot 2}}{2 \cdot 2\sqrt[3]{2}} = 4\sqrt[3]{2}$$

5. $\sqrt{10} \cdot \sqrt{6}$

$$\frac{\sqrt{60}}{\sqrt{4 \cdot 15}} = 2\sqrt{15}$$

6. $\frac{\sqrt[3]{432} \div 16}{\sqrt[3]{128} \div 16} = \frac{\sqrt[3]{27}}{\sqrt[3]{8}} = \frac{3}{2} \text{ or } 1.5$

7. $\sqrt[3]{98} \cdot \sqrt[3]{21}$



$\sqrt[3]{2 \cdot 3 \cdot 7 \cdot 7 \cdot 7} = 7\sqrt[3]{6}$

8. Identify all equivalent expressions to $\sqrt{7} \cdot \sqrt{14}$.

- $\sqrt{98}$ $\sqrt{7 \cdot 14}$ or $(7 \cdot 14)^{\frac{1}{2}}$
- $7\sqrt{7}$ $= \sqrt{98}$ $7^{\frac{1}{2}} \cdot 14^{\frac{1}{2}}$
- $(7 \cdot 14)^{\frac{1}{2}}$ $= \sqrt{2 \cdot 49}$
- $7\sqrt{2}$ $= 7\sqrt{2}$
- $14\sqrt{2}$
- $7^{\frac{1}{2}} \cdot 14^{\frac{1}{2}}$

For questions 9-10, write an equivalent expression by rationalizing the denominator.

9. $\frac{-5\sqrt{3}}{2\sqrt{5}} \cdot \frac{\sqrt{5}}{\sqrt{5}} \rightarrow \frac{-5\sqrt{15}}{2 \cdot \sqrt{5} \cdot \sqrt{5}} \rightarrow \frac{-5\sqrt{15}}{2 \cdot 5} \rightarrow \frac{-5\sqrt{15}}{10} = \frac{-\sqrt{15}}{2}$

10. $\frac{\sqrt{27}}{3\sqrt{6}} \cdot \frac{\sqrt{6}}{\sqrt{6}} \rightarrow \frac{\sqrt{162}}{3 \cdot 6} \rightarrow \frac{\sqrt{81 \cdot 2}}{18} \rightarrow \frac{9\sqrt{2}}{18} \rightarrow \frac{9\sqrt{2} \div 9}{18 \div 9} = \frac{\sqrt{2}}{2}$